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United States: Bank of America Emergency Liquidity Program, 2009¹

*Vincient Arnold*²

Yale Program on Financial Stability Case Study
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Abstract

On December 31, 2008, Bank of America (BoFA) finalized its acquisition of Merrill Lynch, absorbing losses of \$15.5 billion as a result. Regulators were concerned about BoFA's short-term liquidity position and ability to post more collateral if its credit rating was downgraded. On January 16, 2009, the Federal Reserve, Federal Deposit Insurance Corporation (FDIC), and Department of the Treasury announced an interagency support package to BoFA, which included an asset guarantee wherein all three agencies shared losses with BoFA on a "ring-fenced" \$118 billion pool of assets. Under the ring-fencing arrangement, known as the Asset Guarantee Program (AGP), BoFA would absorb the first \$10 billion in losses on the asset pool; the Treasury and FDIC provided \$10 billion in loss protection after that, with BoFA absorbing a further \$1.1 billion. If losses exceeded \$21.1 billion, the Fed agreed to provide a loan to BoFA for 90% of the amount of those losses. The Fed considered the facility an emergency loan and would charge a penalty rate. The Fed's participation was essential to the AGP because it was the only agency that could provide a nonrecourse loan large enough to cover the entire asset pool. This message mattered to the market at the time—it signaled that the government wasn't going to allow the systemic bank to fail. However, based on a stress test analysis of the portfolio by a third-party vendor, the Fed didn't expect it would ever have to extend credit under the facility. BoFA and the authorities never signed definitive documentation for the arrangement, and BoFA never availed itself of the Fed's loan facility or any other component of the AGP. Throughout numerous negotiations, the parties eventually reduced the size of the covered asset pool to \$83 billion. The bank publicly announced on May 7 that it had ended negotiations for the AGP and later agreed to pay a termination fee of \$425 million to the Treasury, Fed, and FDIC, of which the Fed received \$57 million for its lending commitment and operational costs.

Keywords: ad hoc emergency liquidity assistance, Bank of America, Federal Reserve, Global Financial Crisis, Merrill Lynch, ring-fencing

¹ This case study is part of a Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering ad hoc emergency liquidity provision. A survey of all the cases in this series (Kelly et al. 2025) and the individual cases underlying it are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/vol7/iss1/>. Wiggins et al. (2022) surveys broad-based emergency liquidity programs.

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Overview

This case study is about ad hoc emergency liquidity assistance provided to Bank of America³ (BofA) by the Federal Reserve. As part of the broader assistance package, the Department of the Treasury purchased preferred shares in BofA under its Targeted Investment Program (TIP; see Hoffner and Arnold [2024]).

On September 15, 2008—the same day Lehman Brothers filed for bankruptcy—BofA announced that it would acquire Merrill Lynch, then the weakest remaining of the five major US investment banks, which had suffered severe losses during the first three quarters of 2008 (FDIC 2018).

Merrill Lynch’s cumulative net losses for the fourth quarter of 2008 were roughly \$5 billion in mid-November but continued to grow to \$7 billion on December 3, \$9 billion on December 9, and \$12 billion by December 14. Then–BofA CEO Ken Lewis has said the losses were unexpected and that BofA was not aware of the extent of these losses at the time of the shareholder vote to acquire Merrill Lynch on December 5.⁴ On December 17, Lewis met with then–Treasury Secretary Henry M. Paulson, Jr., and informed him that BofA was considering invoking the material adverse change clause, a contractual clause that would allow BofA to renegotiate the merger agreement or leave the deal altogether, in its Merrill Lynch acquisition. The Fed and Treasury expressed concerns that this

Key Terms

Purpose: In support the Asset Guarantee Program and thus to “provide financial support to Bank of America and promote financial stability” (Fed 2009a, 1)	
Announcement Date	January 16, 2009
Operational Date	Not applicable
Termination Date	September 21, 2009
Legal Authority	Section 13(3) of the Federal Reserve Act
Administered By	Federal Reserve Bank of Richmond
Peak Authorization	\$87.3 billion; reduced to \$55.8 billion
Peak Outstanding	AGP, inclusive of Fed residual financing, never effected
Haircut and Recourse	Described by the Fed as nonrecourse, yet BofA would absorb (with the Treasury and FDIC) the first \$21.1 billion in losses and 10% of losses after that. The Fed retained recourse for interest and fees
Interest Rate and Fees	Commitment fee: 20 bps; undrawn: 20 bps; drawn: floating OIS plus 300 bps
Term	Five to 10 years, depending on assets in pool
Part of a Package	The loan facility was accompanied by other Treasury and FDIC loss-sharing agreements and Treasury injections of TARP capital
<i>(continued)</i>	

³ Throughout the case, BofA represents Bank of America Corporation, the bank holding company, often referred to as BAC, rather than Bank of America, N.A., the bank.

⁴ Later, the Securities and Exchange Commission (SEC) would charge BofA with failing to disclose to shareholders before the December 5 vote Merrill Lynch’s known and expected losses—which the SEC said BofA had access to. BofA paid \$150 million to settle the SEC’s enforcement action (FCIC 2011). In the United States, the settlement of a litigation does not imply admission of guilt.

would result in a negative market reaction (FCIC 2011; FDIC 2018).

After agreeing to go forward with the Merrill Lynch acquisition, Lewis and the BofA board of directors asked for a letter committing the government to financial assistance to absorb losses realized in connection with the purchase (SIGTARP 2009). While Secretary Paulson refused, he and then–Fed Chairman Ben Bernanke privately told Lewis that the government would support BofA and “were not going to let a systemically significant institution fail” (SIGTARP 2009, 26).

Outcomes	BofA never signed an agreement to gain access to the loan facility and exited the USG Asset Guarantee Program in May 2009
Notable Features	The loan facility was contingent on large BofA losses to an extent the Fed did not expect to ever lend; the ring-fenced assets would receive a favorable 20% risk weighting, the standard risk weighting for AAA-rated corporate credits under Basel capital standards

On December 31, BofA finalized its acquisition of Merrill Lynch, absorbing \$15.5 billion of Merrill Lynch’s fourth-quarter losses as a result. Government officials became concerned about BofA’s short-term liquidity position⁵ and ability to post additional collateral⁶ if its credit rating were downgraded. Officials were wary that if BofA were unable to meet its obligations, numerous markets would be adversely affected (including those for interbank lending, bank debt, and derivatives), resulting in broad-based runs on depository institutions, expanded repurchase agreement (repo) haircuts, margin calls, and draws on unfunded loan commitments (FDIC 2018).

On January 16, 2009, the Fed, Federal Deposit Insurance Corporation (FDIC), and Treasury announced an interagency support package to BofA consisting of (1) a \$20 billion capital injection of Troubled Assets Relief Program (TARP) funds by the Treasury under TIP and (2) an asset guarantee, wherein the US government shared losses with BofA on a “ring-fenced” \$118 billion pool of troubled assets—the Asset Guarantee Program (AGP). About 75% of the asset pool would come from the former Merrill Lynch portfolio and about 25% would be existing BofA assets (FCIC 2011; FDIC 2018; O’Connor, Bush, and Mayo 2009). The Fed’s term sheet for the BofA package described the eligible assets as, “securities backed by residential and commercial real estate loans and corporate debt, derivative transactions that reference such securities, loans, and associated hedges, as agreed, and such other financial instruments as the U.S. government (USG) has agreed to guarantee or lend against” (Fed 2009a, 6). At the time, the parties had not determined the specific assets that would be included in the asset pool; the details would be settled if and when the parties signed an agreement. The asset pool would exclude foreign assets, equity instruments, and securities issued or originated after March 14, 2008. The US government maintained the right to refuse any assets from inclusion in the asset pool (Fed 2009b).

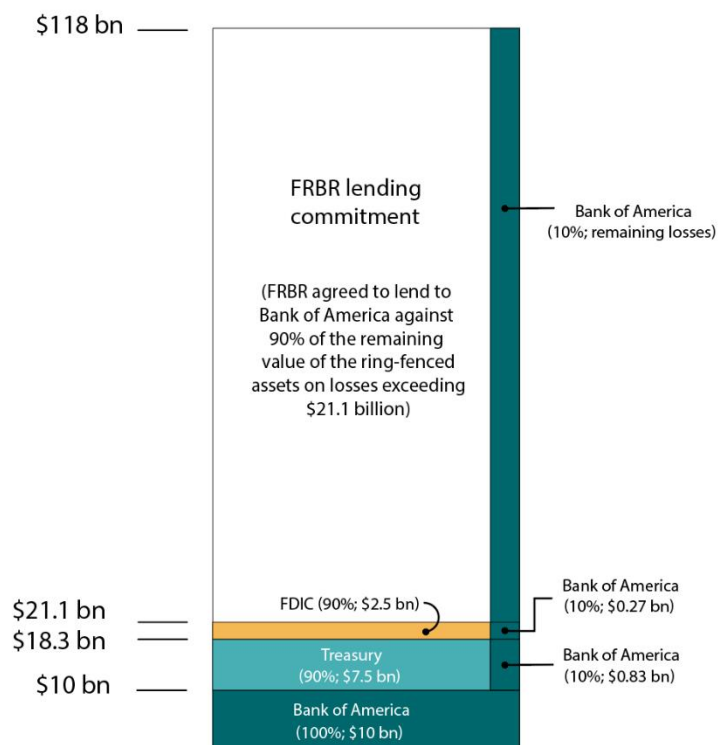
⁵ For example, 67% of BofA’s repurchase agreement (repo) and securities-lending funding (\$384 billion, or 23.4% of BofA’s total liabilities) was rolled every night (BofA 2009b; FCIC 2011).

⁶ For example, a one-notch downgrade would require BofA to post an extra \$10 billion in collateral to market counterparties (FCIC 2011).

The first \$21.1 billion in losses on the asset pool would be shared by BofA, the Treasury, and the FDIC, according to the breakdown shown in Figure 1. The Fed would provide an emergency loan for 90% of the amount of any losses over \$21.1 billion through a nonrecourse loan facility at the Federal Reserve Bank of Richmond (FRBR) (FDIC 2018). Figure 1 shows the Fed's loss position in the broader asset guarantee structure.

The Fed described the loan facility as a "residual financing." In the event losses exceeded \$21.1 billion, the Fed committed to lend against the remaining \$97 billion of assets in the asset pool, through the FRBR (whether the lending would be in the form of a one-time lump-sum loan or numerous loans remains unclear; see Key Design Decision No. 7, Source and Size of Funding). That lending would be collateralized against the asset pool, with recourse to BofA only for interest and fees. BofA would remain liable for 10% of the remaining losses, with the Fed lending 90% of its asset pool coverage. Based on available documentation, it appears that BofA would also pay the Fed interest on the entire outstanding amount of the loan (90% of the Fed's asset pool coverage) as long as it was in place (Treasury 2009).

A third-party vendor, Pacific Investment Management Company (PIMCO), projected losses would reach only \$20 billion in a stressed scenario. Given that the loss-sharing agreement among BofA, Treasury, and FDIC covered more than that amount, the Fed didn't expect it would be required to extend any credit under the facility (Glassman et al. 2009). According to Jason Cave, an FDIC official party to the government negotiations, the Fed's participation in the AGPs for BofA and Citigroup through residual financing facilities was primarily for the announcement or signaling effect; the residual financing facilities were not strictly necessary in a financial sense because of the low likelihood the banks would use them (Cave 2024).

Figure 1: Bank of America Loss Protection Positions

Note: The FDIC and Treasury, while sharing the second-loss position losses asymmetrically (75% of losses to Treasury and 25% to FDIC), were both in a second-loss position on a *pari passu* basis. In other words, for a given \$10 billion loss in excess of the first \$10 billion, BofA would absorb \$1 billion, and the remaining \$9 billion would be shared 75% to Treasury (\$6.75 billion) and 25% to the FDIC (\$2.25 billion).

Sources: FDIC 2018; Fed 2009a.

On May 7, 2009, BofA announced that it was ending negotiations with the government over the Asset Guarantee Program (BofA 2009a; Treasury 2009). BofA and the authorities had never signed definitive documentation for the Fed's loan facility, and BofA had never availed itself of the loan facility or any other component of the Asset Guarantee Program (Treasury 2009; Treasury n.d.). BofA did, however, receive capital injections through TIP, the other major component of the BofA package (see Hoffner and Arnold [2024]). After having agreed to reduce the size of the asset pool to \$104 billion, the parties were unable to reach agreement on the definition of eligible losses for \$42 billion of those assets. As a compromise, they reduced the asset pool by half that amount, \$21 billion, resulting in an adjusted size of \$83 billion for the asset pool (Treasury 2009).

BofA and the authorities agreed that it would be appropriate for BofA to compensate, on a prorated basis, the government for its "willingness to bear the risk of loss on the pool during the period from the date of the term sheet to May 6, 2009" (Treasury 2009). On September 21, 2009, BofA agreed to pay a termination fee of \$92 million to the FDIC, \$276 million to Treasury, and \$57 million to the Fed for its commitment to lend against the asset pool and its operational costs (Treasury 2009). For a discussion of BofA's exit from its TIP investments

from the Treasury, see Hoffner and Arnold (2024). Figure 2 shows a timeline of the key events in the BofA AGP intervention.

Figure 2: Timeline of Bank of America Asset Guarantee Program

Date	Event
Sep. 15, 2008 (Mon.)	BofA announces intent to acquire Merrill Lynch; Lehman Brothers fails.
Oct. 28, 2008 (Tues.)	Treasury uses \$15 billion of TARP capital to buy preferred shares of BofA and commits to purchase \$10 billion of preferred shares of Merrill Lynch with settlement deferred pending BofA merger.
Nov. 26, 2008 (Wed.)	The Fed approves the BofA and Merrill Lynch merger, says both banks are well capitalized and that BofA “has sufficient financial resources” to complete the transaction.
Dec. 17, 2008 (Wed.)	BofA informs Treasury that it is considering invoking the material adverse change clause in the Merrill Lynch merger agreement as a result of larger-than-expected losses at Merrill Lynch.
Dec. 31, 2008 (Wed.)	BofA acquires Merrill Lynch.
Jan. 9, 2009 (Fri.)	Fed and Treasury officials approach the FDIC regarding the possibility of ad hoc assistance to BofA.
Jan. 15, 2009 (Thurs.)	- Fed Board of Governors and the FDIC board of directors recommend that the secretary of the Treasury invoke the systemic risk exception. - Board of Governors of the Fed decides to provide BofA financing support.
Jan. 16, 2009 (Fri.)	- Fed, Treasury, and FDIC announce interagency support package consisting of TARP capital injections and asset guarantees, including the Fed’s loan facility for loss-sharing on the pool of assets under guarantee. - BofA discloses \$15.5 billion Merrill Lynch loss on Q4 earnings call.
May 7, 2009 (Thurs.)	BofA announces its intention to terminate the USG aid package after concluding that losses would not exceed its \$10 billion first-loss position.
Sep. 21, 2009 (Mon.)	BofA and government agencies sign termination agreement.

Source: FCIC 2011; author's analysis.

Summary Evaluation

On January 16, 2009, BofA’s stock closed down 14% on the day, despite the Asset Guarantee Program—of which the Fed’s conditional loan facility was just the most senior part—when the USG announced the package and BofA disclosed its Merrill Lynch–related losses (FCIC 2011).

Nonetheless, in an evaluation of the AGP, the Treasury noted that:

Although the agreement was never implemented, its initial announcement (and the announcement of the Citigroup transaction [. . .]) was widely welcomed by the markets and contributed immediately to helping restore investor confidence in these financial institutions and the banking system generally. (Treasury n.d.)

During BofA's Q4 2008 earnings call on January 16, 2009, a NAB Research analyst said that BofA had been referred to as a "ward of the state" for the USG support package and referred to the aid as "government, quote, investments" (O'Connor, Bush, and Mayo 2009, 13). UBS analysts said that the Asset Guarantee Program would reduce losses and improve liquidity but that it remained "unclear how this [the AGP and TIP capital] will rebuild meaningfully without significant common equity issuance over time" (Placet and O'Connor 2009, 1). Also on the call, BofA CEO Ken Lewis said that BofA would exit the Asset Guarantee Program "as soon as possible" (O'Connor, Bush, and Mayo 2009, 13).

Chairman Bernanke said of the Fed's support to BofA during its acquisition of Merrill Lynch: "I think I have nothing that I regret about the whole transaction. I think it was, in fact, a very successful operation overall and it achieved the public policy objectives that were very important" (US House Oversight Committee 2009b, 39).

Former FDIC official Jason Cave, who was involved in structuring the AGP, told YPFS that the asset guarantee programs were not worth doing, as it was far too complicated to be considered an optimal policy response. In his view, the main value of the Fed's participation in the AGP was the announcement or signaling effect with the public and its sister agencies (Cave 2024).

Context: United States 2009–2010	
GDP (SAAR, nominal GDP in USD)	\$14.5 trillion in 2009
	\$15.0 trillion in 2010
GDP per capita (SAAR, nominal GDP in USD)	\$47,195.0 in 2009
	\$48,650.7 in 2010
Sovereign credit rating (long-term debt)	Moody's: Aaa in 2009
	S&P: AAA in 2009
	Fitch: AAA in 2009
	Moody's: Aaa in 2010
	S&P: AAA in 2010
	Fitch: AAA in 2010
Size of banking system	\$9.1 trillion in 2009
	\$9.2 trillion in 2010
Size of banking system as a % of GDP	62.7% in 2009
	60.8% in 2010
Size of banking system as a % of financial system	28.5% in 2009
	28.2% in 2010
Five-bank concentration of banking system	44.3% in 2009
	46.1% in 2010
Foreign involvement in banking system	19% in 2009
	16% in 2010
Existence of deposit insurance	Yes in 2009
	Yes in 2010

Sources: Bloomberg; World Bank Deposit Insurance Dataset; World Bank Global Financial Development Database.

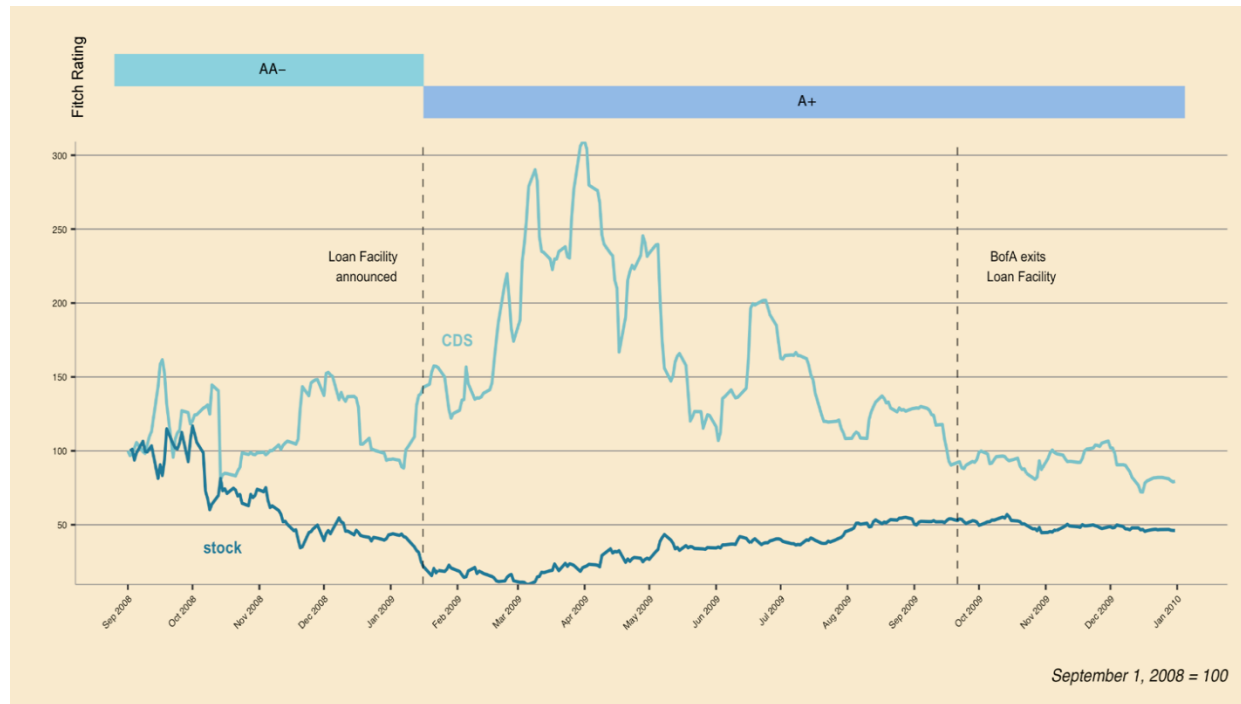
Key Design Decisions

- 1. Purpose: The Fed created the loan facility as a supporting component of the Treasury's Asset Guarantee Program, a loss-sharing arrangement for BofA, to restore confidence in BofA and support financial stability.**

As of September 30, 2008, BofA was the largest banking organization in the United States with \$2.7 trillion in assets. Its primary bank subsidiary, Bank of America, N.A., was the second-largest bank in the country. The bank held more than 10% of US bank deposits, had total consolidated assets of \$1.4 trillion, and was the largest issuer of insured deposits. BofA was a major supplier of credit to the US economy and, following its merger with Merrill Lynch, the largest securities broker in the world. In addition to deposits, the combined BofA organization had \$426.3 billion of debt outstanding in commercial paper, other short-term debt, and long-term debt. BofA was also a counterparty to numerous financial institutions throughout the world (BofA 2009b; FDIC 2018; Fed 2009a).

According to Fed Chairman Bernanke, the Fed's decision to assist BofA through its loan facility as part of the Asset Guarantee Program occurred "under highly unusual circumstances in the face of grave threats to [the] financial system and [the] economy" (US House Oversight Committee 2009b, 19). Against the backdrop of the Global Financial Crisis of 2007–2009 (GFC), BofA was negatively impacted by market conditions both directly and through Merrill Lynch upon its acquisition. On October 28, the Treasury had purchased \$15 billion of BofA preferred stock using TARP funds (Fed 2009a). By year-end 2008, BofA's low levels of tangible common equity had worried the market (FCIC 2011). At the beginning of 2009, BofA's stock price had fallen 70% in 2008, and BofA was preparing to announce Q4 2008 results below market expectations (FDIC 2018). BofA's stock price was declining, and the cost of insuring against the bank's failure was rising (see Figure 3).

Figure 3: Bank of America: Stock Prices and Credit Default Swap Spreads (Scaled) with Credit Ratings, 2008–2009



Sources: Fitch Ratings; LSEG/Refinitiv; author's calculations.

In December 2008, government officials were in communication with BofA about potential government assistance necessary to support BofA because of the firm's assumption of losses from Merrill Lynch (FCIC n.d.). On January 16, 2009, BofA announced \$1.8 billion in losses (not inclusive of losses resulting from its acquisition of Merrill Lynch)—its first quarterly loss in 17 years (Stempel 2009). The Fed said that, without a support package, the market reaction to BofA's earnings could result in contagion:

These losses had the potential to weaken materially investor and counterparty confidence in the combined organization and to hamper the ability of the organization and its insured depository institutions to continue to obtain funding in the currently fragile credit markets. Given current market conditions, such adverse developments at the organization, if left unaddressed, could have resulted in other financial institutions experiencing similar funding problems, posed risks to financial stability, and increased downside risks to economic growth. (Fed 2009a, 2–3)

The BofA program was modeled on the previous guarantee program that the authorities had created for Citigroup. Policymakers employed the ring-fencing strategy because it convinced markets that the government had taken extreme risk (“the left tail”) off of the bank's balance sheet, was easily replicable, and would cost the government less than other strategies, such as further capital injections, asset purchases, or conservatorship. For a detailed discussion of the Citigroup Asset Guarantee Program, see Arnold (2025).

In response to market pressures on BofA and its systemic importance, the Fed approved a loan facility to BofA for 90% of the amount of mark-to-market and credit losses exceeding \$21.1 billion accrued on a specified pool of assets (Fed 2009a). The Federal Reserve Bank of Richmond, which was to operationalize the loan facility, did not expect to lend under the BofA program given the size of the loss absorption provided by the Treasury and then the FDIC (GAO 2011).

However, the Fed's participation was needed in the Asset Guarantee Program's structure because it was the only agency that could provide a loan large enough to cover the entire asset pool. According to the Congressional Oversight Panel:

The TARP purchasing authority [was] reduced dollar-for-dollar by the amount guaranteed, meaning that insuring an asset under Section 102 of [the Emergency Economic Stabilization Act of 2008 had] almost an equivalent impact on TARP purchasing authority as purchasing the same asset. Treasury *needed the joint participation of the Federal Reserve* and the FDIC to cover the sizeable Citigroup and Bank of America guarantees. While the Federal Reserve would provide financing only after the loss sharing agreements with Treasury and the FDIC were exhausted, *it [was] the only agency that could provide a non-recourse loan of large notional value, if necessary, because of its emergency lending authority under Section 13(3) of the Federal Reserve Act.* (COP 2009, 39) [emphasis added; COP citations omitted]

In other words, using TARP to cover the full amount of the Asset Guarantee Program would have burned through more than 15% of the entire \$700 billion Congress had allocated to Treasury under the program.

On the other hand, the Fed's residual financing facility was not strictly necessary in a financial sense, according to Jason Cave, an official at the FDIC involved in the BofA AGP. In his view, the main value of the Fed's participation in the AGP was the announcement or signaling effect with the public and its sister agencies.

Regulators planned the assistance package to BofA in advance of its scheduled January 16 earnings call to preempt a negative market response (FDIC 2018).

Other Options

The government also considered as other options a government backstop of a capital raise, a government injection of common equity with limited control rights, and a capital offering with a ring-fence (instead of a government capital injection with a ring-fence) (BofA 2008; FCIC n.d.). FDIC regulators considered other options like a covered bond structure for BofA, but the Treasury and Fed thought that they would get more for each dollar of TARP funding by using a guarantee structure (Cave 2024).

2. Part of a Package: The Fed loan facility facilitated the Asset Guarantee Program and was part of a package that also included a capital injection from the Treasury.

Under the Asset Guarantee Program, the Treasury and FDIC agreed to guarantee ring-fence-eligible assets by taking second position on a *pari passu* basis, 75% to the Treasury and 25% to the FDIC. Under this arrangement, BofA would bear the first \$10 billion in losses on the asset pool. After that, BofA would bear 10% of the second-loss layer, with and Treasury the FDIC providing the other 90% of the second-loss layer. While the Treasury and FDIC would absorb those losses on a *pari passu* basis, 75% of the loss guarantee would be covered by the Treasury and 25% by the FDIC. In compensation for their loss protection, BofA would issue the Treasury and FDIC \$3 billion and \$1 billion, respectively, in preferred stock bearing an 8% coupon. Treasury also received warrants to purchase common stock with an aggregate exercise value of 10% of the total amount of preferred issued (Fed 2009a).

At the same time as it agreed to the AGP, Treasury agreed to purchase \$20 billion in senior preferred stock bearing an 8% coupon with TARP funds through the Targeted Investment Program (FDIC 2018). BofA received the TIP capital injection on January 16, 2009. As required by the TARP enabling legislation, the Treasury also received warrants for the purchase of common stock at a strike price of \$13.30 per share, for an aggregate value of \$2 billion (SIGTARP 2009). (For details on the TIP, see Hoffner and Arnold [2024].)

BofA also utilized several other broad-based crisis response programs. It had already borrowed substantial amounts from the Fed's Term Auction Facility (an auction-based discount window liquidity facility), the Primary Dealer Credit Facility, the Term Securities Lending Facility, and the Commercial Paper Funding Facility (FCIC 2011; Fed 2016). It also issued \$31.7 billion in FDIC-guaranteed senior debt under the Temporary Liquidity Guarantee Program. In addition to utilizing these crisis programs, BofA borrowed \$92 billion from the Federal Home Loan Bank System, which comprises government-sponsored enterprises (FCIC 2011).

Pursuant to the term sheet, regulators would have agreed to assign the ring-fenced assets a favorable 20% risk weighting, the standard risk weighting for AAA-rated corporate credits under Basel capital standards (BIS 2004; Fed 2009a). Although the company did not release details, the troubled assets in the asset pool would likely have carried risk weightings of closer to 80% based on the information the bank did provide.⁷ The favorable regulatory treatment would have lowered BofA's capital requirements on the ring-fenced assets by from about \$7.5 billion to \$1.9 billion.⁸ BofA said that the risk-weighting adjustment as a result of the Asset Guarantee Program would have lowered its risk weighted assets by \$70 billion. The regulators' decision to grant a 20% risk weight to the asset pool thus appears to represent significant regulatory forbearance—they were allowing BofA to hold just \$1.9 billion in capital against an asset portfolio on which BofA remained exposed to the first

⁷ BofA said the Asset Guarantee Program lowered its risk-weighted assets by \$70 billion. The 20% risk-weighting on the \$118 billion asset pool is \$24 billion. So, their original risk-weighted assets were roughly \$94 billion (\$70 billion + \$24 billion), or about 80% of \$118 billion.

⁸ The standard 8% capital charge would result in a \$7.5 billion capital requirement based on \$94 billion in risk-weighted assets and \$1.9 billion based on \$24 billion in risk-weighted assets (O'Connor, Bush, and Mayo 2009).

\$10 billion in losses. BofA said that the BofA package—including the \$20 billion in TIP, the \$4 billion fee paid in preferred shares, and the regulatory relief—would have improved its Tier 1 capital ratio on a pro forma basis from 8.7% to 10.7% (O'Connor, Bush, and Mayo 2009).

The Fed's role, through the loan facility, was not specifically necessary for this risk-weighted asset treatment. It was not strictly necessary in a financial sense either, according to Cave at the FDIC, since nobody expected the bank to exercise the loan. In Cave's view, the main value of the Fed's participation in the AGP was the announcement or signaling effect with the public and its sister agencies (Cave 2024).

3. Legal Authority: Section 13(3) of the Federal Reserve Act provided the legal authority for the Fed to create and lend through the loan facility.

Section 13(3) of the Federal Reserve Act (FRA, 12 U.S.C. § 343) provided the legal authority for the Fed to create and lend through the loan facility. The Fed's Board of Governors voted to invoke Section 13(3) on January 15, 2009 (Fed 2009a). Section 13(3) permitted the Fed, "in unusual and exigent circumstances," to provide liquidity "for any individual, partnership, or corporation," provided the assistance was "secured to the satisfaction" of the Fed (US Congress 2008, sec. 13[3]). The invocation of Section 13(3) allows the Fed to provide liquidity more broadly than its monetary policy and discount window authorities allow (US Congress 2008, secs. 10B, 13, 14).⁹

Fed legal staff and policymakers consistently interpreted Section 13(3)'s "secured to the satisfaction" requirement as meaning that the Fed had to be reasonably confident of full repayment. In other words, while the Fed might take some losses in 13(3) lending, in order to provide liquidity under Section 13(3), it had to make the ex ante determination that it would be repaid (Alvarez 2022). According to the GAO, the Fed didn't expect it would be required to extend any credit under the facility, based on the analysis of a third-party vendor projecting that losses would not exceed \$20 billion in a stressed scenario (less than the amount that BofA, the FDIC, and the Treasury had committed to cover before the Fed facility would be required).

Given that the BofA Asset Guarantee Program had a maturity of five to 10 years (see Key Design Decision No. 9, Loan Duration), it was possible that the Fed's loan facility lending commitment could have been activated after the "unusual and exigent circumstances" required by Section 13(3) had passed (US Congress 2008a, sec. 13[3]). The Fed's view on this, according to then-General Counsel Scott Alvarez, was that if the ring-fenced losses had

⁹ As noted in Overview and Key Design Decision No. 10, Balance Sheet Protection, the asset pool contained a significant amount of derivative assets. Section 13(3) authorizes the Fed to provide liquidity only against note collateral—that is, instruments of credit—a classification that would typically not apply to derivatives. While the loan facility would technically be a loan to BofA, its nonrecourse status generally would negate that fact. However, this legal risk would be mitigated by the Fed's recourse to BofA for interest and fees; the junior protection from BofA, Treasury, and the FDIC; and the Fed's "incidental" authorities under Section 4(4) of the FRA, as it could be argued that taking the derivative collateral functioned as incidental to the Fed's 13(3) authority. See (Alvarez 2022; Alvarez et al. 2008; Kelly 2023).

penetrated through the junior layers to the point where the Fed had to lend into the ring-fence, that situation itself would satisfy the “unusual and exigent circumstances” requirement (Alvarez 2022).

4. Administration: The Fed’s loan facility was set up by the FRBR and would have been administered by the FRBR had BofA accessed it.

The Fed’s loan facility, to the extent that it was a constituent part of the broader BofA package, was the product of “significant and intense” intragovernmental negotiations involving the Treasury, FDIC, and Office of the Comptroller of the Currency in January 2009 surrounding key terms of the package, including duration, scope, and fees (US House Oversight Committee 2009b, 19). According to Jason Cave, then at the FDIC, the BofA AGP was significantly more complicated to design than its Citi predecessor because of the opacity of the assets it had at the time (see Figure 8 at Appendix) (Cave 2024).

While the Fed never lent under the loan facility (see Key Design Decision No. 9, Loan Duration), the FRBR was the designated administrator of the facility (Fed 2009a).¹⁰ The FRBR never finalized an agreement with BofA to provide the loan facility (GAO 2011). The GAO referred to the BofA lending commitment as an “agreement-in-principle” (GAO 2011, 185).

On January 9, the FRBR engaged PIMCO to provide valuation services in connection with the loan facility for a fee of \$12 million. There was no competitive bidding process because of the time constraints. On January 16, the FRBR also engaged Ernst & Young to conduct due diligence in connection with the loan facility for a fee of \$10.6 million (without a competitive bidding process). While the FRBR paid all vendor fees, BofA was required to reimburse the FRBR for those fees (GAO 2011).

BofA was eligible to draw on the loan facility if and when it accrued losses (unrealized mark-to-market losses and realized credit losses) on the asset pool exceeding \$21.1 billion. BofA could terminate the loan facility commitment and prepay any outstanding loans at any time, subject to the Fed’s approval (Fed 2009b).

5. Governance: The GAO conducted a review of the Fed’s loan facility as part of its broader review of the US government’s assistance to BofA; Fed Chairman Bernanke testified to Congress about the BofA package.

Section 129 of the EESA, passed on October 3, 2008, required the Fed to report to the Committee on Banking, Housing, and Urban Affairs of the Senate (Senate Banking Committee) and the Committee on Financial Services of the House of Representatives (House Financial Services Committee) on any use of its Section 13(3) authority within seven days of invoking that authority (see Key Design Decision No. 6, Communication and Disclosure) (US Congress 2008b, sec. 129[a]).

¹⁰ Bank of America is headquartered in Charlotte, North Carolina, which is in the FRBR’s district.

In July 2011, the GAO published a report on the Fed's emergency programs, including an appendix specifically on the BofA loan facility ("agreement in principle") (GAO 2011). The GAO's findings were that:

- FRBR effectively implemented conflicts of interest controls in its BofA program, for example excluding BlackRock as a service provider due to BofA's significant stake in BlackRock; and
- FRBR did not use a competitive procurement process to award contracts to the private sector entities hired to provide services (see Key Design Decision No. 4, Administration) (GAO 2011).

The GAO report did not include any further commentary on the Fed decision-making in its implementation of the loan facility (GAO 2011).

On June 11 and June 25, 2009, Chairman Bernanke testified before a congressional House committee about the Fed's involvement in BofA's acquisition of Merrill Lynch (US House Oversight Committee 2009a; US House Oversight Committee 2009b).

6. Communication and Disclosure: Both BofA and the Fed disclosed the loan facility in annual reports; the Fed made a disclosure to Congress pursuant to legal requirements. The Fed also opted to release it to the public.

On January 16, 2009, the Fed issued a joint press release with the Treasury and FDIC announcing the BofA package. The press release said that, "if necessary, [the Fed stood] ready to backstop residual risk in the asset pool through a non-recourse loan" (Fed, FDIC, Treasury 2009). Policymakers timed this announcement to try to head off the adverse market reaction they expected to BofA's announcement of its Merrill Lynch losses (FDIC 2018). The press release said that the USG's support package to BofA was necessary to protect US taxpayers and strengthen the financial system (Fed, FDIC, Treasury 2009). On the same day, BofA held its fourth-quarter 2008 earnings call, in which then-CEO Ken Lewis referred to the Asset Guarantee Program (though he did not explicitly mention the Fed's loan facility) and the assurance that it gave to BofA to move forward with the Merrill Lynch acquisition on its original terms (O'Connor, Bush, and Mayo 2009).

The *Financial Times* reported the BofA package after the early-morning announcement. The article described the BofA Asset Guarantee Program as similar to the Citigroup Asset Guarantee Program, which occurred on January 15, 2009 (see Arnold [2025]). Before the announcement, rumors had been spreading that a large US bank would be nationalized (Tucker 2009).

In its 2009 annual report, the Fed disclosed that in September 2009, BofA paid a fee to exit the arrangement (Fed 2010).

BofA announced its intention to terminate the AGP on May 7, the same day it announced its results in the Fed's stress test. According to Joe Price, then the chief financial officer, in that statement:

We believe that the expense of the asset wrap exceeds the potential benefit, especially since, even under our stress test conclusions, losses never exceed the initial \$10 billion we would have to cover . . . We have already taken substantial action on our own to reduce exposure to the covered assets while incurring minimal losses. (BofA 2009a)

In its 2009 annual report, BofA disclosed that it had reached an agreement to terminate the term sheet for the agreement-in-principle with the USG; it did not provide any details on the loan facility or the USG Asset Guarantee Program (BofA 2010).

Section 129 of the EESA required the Fed to report to the Senate Banking Committee and House Financial Services Committee on any use of its Section 13(3) authority within seven days of invoking that authority, including (1) its justification for invoking Section 13(3) and (2) the specific terms of any Fed actions, specifically, “the size and duration of the lending, available information concerning the value of any collateral held with respect to such a loan, the recipient of warrants or any other potential equity in exchange for the loan, and any expected cost to the taxpayers” (US Congress 2008b, sec. 129[a][2]). Pursuant to its Section 129 reporting requirements, the Fed issued a Section 129 report on its loan facility on January 15, 2009 (before its announcement to the public) (Fed 2009a). In its Section 129 Disclosure, the Fed said that the AGP—of which its loan facility was part—would “help restore confidence” in BofA and would “promote financial stability” (Fed 2009a, 3). The Fed also said that it did not expect its loan facility to result in any losses to the Fed or the taxpayer and presented other disclosures required by Section 129 (subsection [a][2]). The Section 129 Disclosure included the term sheet, which provided a broad description of the assets eligible for guarantee but stipulated that no assets would be assigned to the asset pool until BofA and the USG had agreed to effectuate the AGP.

The Fed made its January 15, 2009–dated Section 129 Disclosure publicly available on February 18, 2009. The FDIC's January 15, 2009–dated memorandum was made publicly available upon the publication of the Financial Crisis Inquiry Commission Report in 2011.

7. Source and Size of Funding: The Fed would have funded the loan facility through the creation of central bank reserves and up to 90% of covered losses.

The Fed approved the loan facility to BofA for 90% of mark-to-market and credit losses exceeding \$21.1 billion accrued on a \$118 billion specified pool of assets (Fed 2009a). The Fed funds liquidity provision through the creation of central bank reserves.

After the junior layers were exhausted (\$10 billion BofA layer and \$11.1 billion split 90% to the FDIC/Treasury and 10% to BofA), the Fed would have lent BofA 90% of \$97 billion (\$87.3 billion) to cover the remaining value of the asset pool, assuming the asset pool's size didn't change. BofA would have been responsible for reimbursing the Fed for 10% of any losses on the loan facility's portion of the asset pool (GAO 2011).

The size of the loan or loans that the Fed would extend to BofA once losses passed the \$18 billion threshold isn't clear from the documentation. The documentation for the Citi package, in contrast, states clearly that the Fed would have immediately extended a loan equal to the entire remaining value of the asset pool as soon as its threshold was reached. While

authorities typically described the BofA deal as mirroring the Citi deal, it's possible that they planned to structure the BofA deal slightly differently, allowing BofA to borrow multiple tranches when losses occurred. The initial term sheet for the BofA transactions leaves open that possibility. It says that BofA could “draw on [the] Federal Reserve loan facility” when the threshold was passed, and it says the Fed’s fee would be based on the undrawn amounts (Fed 2009b). Moreover, Citi’s term sheet refers to “a non-recourse *loan*,” while BofA’s term sheet refers to “any Federal Reserve *loans*” (Fed 2008a; Fed 2009b) [italics added].

In negotiations to finalize and execute the term sheet, the USG and BofA subsequently reduced the asset pool from \$118 billion to \$104 billion (which would have reduced the Fed’s share to \$83 billion). Thereafter, the parties were unable to reach agreement on the definition of eligible losses for \$42 billion of those assets. As a compromise, they reduced the asset pool by half that amount, \$21 billion, resulting in an adjusted size of \$83 billion for the asset pool. The Fed’s ultimate loan commitment was \$55.8 billion after the \$21 billion loss-sharing by BofA, Treasury, and the FDIC, and BofA’s 10% loss buffer (Treasury 2009). Figure 4 shows the evolution of the asset pool size from its original announced size to its size at termination.

Figure 4: Bank of America Asset Guarantee Pool and Federal Reserve Loan Sizes (\$ billions)

	Original	Negotiated	Final Termination
(a) Asset Pool Size	118	104	83
(b) Fed Exposure ¹	97	83	62
(c) Fed Loan Size ²	87.3	74.7	55.8

¹ equal to (a) – 21

² equal to (b) * 0.9

Sources: Fed 2009b; Treasury 2009.

8. Rates and Fees: If the loan agreement had been effected, the Fed would have imposed a fee of 20 basis points (bps) on undrawn amounts of the loan facility and charged three-month OIS plus 300 bps on drawn amounts; the Fed levied a commitment fee.

The Fed would have charged 20 bps on all undrawn amounts, had the parties finalized the agreement (Fed 2009b). Assuming the Fed’s ultimate maximum exposure was \$62 billion, less the 10% BofA loss-sharing (for an ultimate loan size of \$55.8 billion), this would have implied an annual fee of \$111 million (Treasury 2009).

If BofA had accessed the loan facility, the Fed would have charged a floating rate of three-month Overnight Index Swap (OIS) rate plus 300 bps on the entire amount of the facility (Fed 2009b). Based on the Fed’s commitment fee calculation, that amount would have been the Fed’s ultimate loan size (equal to 90% of the Fed’s overall loss layer, ultimately \$55.8 billion—see Figure 5 row [c]) (Treasury 2009). The OIS rate is a market-determined rate representing expectations for the Fed’s policy rate over different tenors—in this case, three months. At the time, the Fed’s “Regulation A”—the regulation it wrote to guide its

implementation of Section 13(3) of the Federal Reserve Act—called for Fed emergency liquidity to carry an interest rate “above the highest rate in effect for advances to depository institutions”—a reference to the Fed’s discount window (Fed 2008c, vol. Code of Federal Regulations, title 12, sec. 201.4). At the time, the discount window’s primary credit program charged the Fed’s effective policy rate plus 35 bps; the rarely used secondary credit program was priced at the policy rate plus 85 bps (Fed 2023; FRBSL 2023).

BofA ultimately terminated the Asset Guarantee Program before the parties finalized it. Upon the termination of the arrangement, BofA agreed to compensate the government on a pro-rated basis for its “willingness to bear the risk of loss on the Pool during the period from the date of the term sheet to May 6, 2009” (Treasury 2009). On September 21, 2009, BofA agreed to pay a termination fee of \$425 million to the government.

The Fed’s portion of the \$425 million fee was \$57 million. That included \$34 million for its commitment to lend against the pool and \$23 million to cover Fed operational expenses. The \$34 million represented the pro rata amount of the 20 bps fee for the 110 days in which the negotiations took place from January 16 to May 6; it was charged on the net Fed loan exposure of \$55.8 billion (which represented 90% of the Fed’s overall loss layer—see Figure 5) (Treasury 2009). The Citigroup Asset Guarantee Program—on which the BofA ring-fence was designed—did not feature a commitment fee (Arnold [2025]).

9. Loan Duration: The loan facility would have been available to BofA for the duration of the government’s asset guarantee—up to 10 years, depending on the type of assets guaranteed. BofA exited the Asset Guarantee Program on May 6, 2009.

The loan facility’s maturity schedule would have aligned with the durations of the asset guarantees: as much as 10 years for residential mortgage-related assets and five years for other assets (Fed 2009a).

On May 6, 2009, BofA expressed interest in terminating the assistance package. BofA and the government had never signed definitive documentation for the arrangement, and BofA had never availed itself of the loan facility or any other component of the Asset Guarantee Program. However, BofA agreed to compensate the USG for the period between the date of the term sheet (January 15) and May 6. BofA and the USG signed a termination agreement on September 21, 2009 (Treasury 2009; Treasury n.d.).

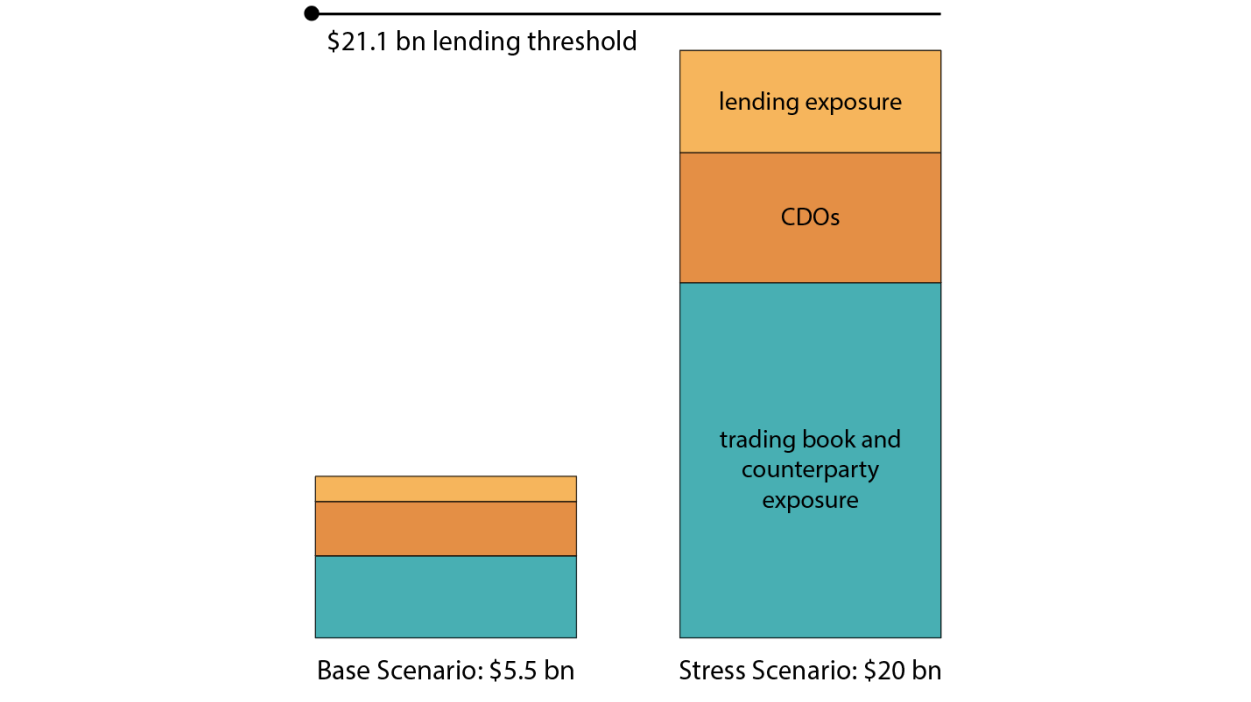
According to the Fed, there were two primary reasons that BofA decided to terminate the Asset Guarantee Program. First, BofA wanted loss coverage for unrealized as well as realized losses, and the lack of protection for mark-to-market losses was unappealing to BofA; while the Fed’s loan facility considered mark-to-market losses eligible for the determination of lending size, the Treasury and FDIC did not—their guarantee was only for realized losses. It is possible that this provision was still under negotiation between BofA and the government parties (or between the government parties internally) when the parties terminated the program. Second, BofA was one of the financial institutions subject to the ongoing Supervisory Capital Assessment Program (SCAP), the stress test exercise that the Fed had

started in February 2009—so both parties had agreed to delay negotiations on finalizing the Asset Guarantee Program until after the test was completed in May. The SCAP was a stress test for the largest banks in the country to determine which banks would need more capital to withstand a severe downturn. The result of the SCAP showed that BofA had a \$33.9 billion capital shortfall; however, the bank was able to raise \$35.9 billion in capital on private markets through asset sales, share conversions, and the issuance of new shares over eight months. Based on the SCAP results, BofA decided the Asset Guarantee Program was not sufficiently valuable to finalize (GAO 2011; Lawson 2021; NYT 2009).

10. Balance Sheet Protection: While the loan facility was to be extended to BofA on a nonrecourse basis, it was collateralized by the asset pool under the Asset Guarantee Program and protected by the commitment of BofA, FDIC, and Treasury to cover the first \$21.1 billion of losses. The Fed retained recourse for interest and fees on the facility, and the interest on undrawn and drawn amounts also mitigated the Fed’s exposure.

The FRBR never expected to lend through the loan facility. The Fed would not have to lend until BofA made \$21.1 billion in losses, but an independent analysis from PIMCO showed that, even under a stress scenario, expected losses on the asset pool did not exceed the \$21.1 billion mark above which FRBR would be obligated to lend through the loan facility; see Figure 5 (GAO 2011; Glassman et al. 2009).

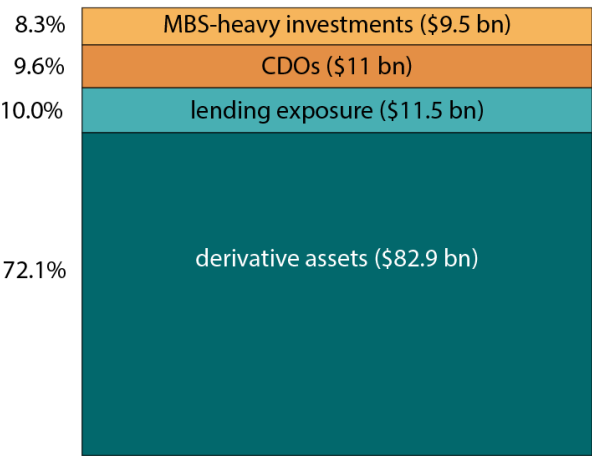
Figure 5: PIMCO-Generated Base and Stress Loss Scenarios, Bank of America



Sources: Glassman et al. 2009; author’s analysis.

The Fed’s loan facility lent against the asset pool, which itself comprised mostly (about three-quarters) Merrill Lynch assets and some (about one-quarter) legacy BofA assets (Fed 2009b; O’Connor, Bush, and Mayo 2009). The government excluded four classes of assets from the pool: (1) foreign assets; (2) assets originated (or issued) after March 14, 2008; (3) equity assets; and (4) any other assets deemed necessary to exclude at the discretion of the USG (Fed 2009b). The guaranteed asset pool primarily comprised derivative assets (inclusive of hedged collateralized debt obligations) (Glassman et al. 2009).¹¹ Figure 6 shows the BofA asset pool in December 2008. For PIMCO’s valuation of the proposed asset pool, see Figure 7 at Appendix.

Figure 6: Bank of America, Ring-Fencing Asset Pool, December 31, 2008



Sources: Glassman et al. 2009; author’s analysis.

The loan facility was designed to be extended on a nonrecourse basis,¹² meaning that the Fed had no claim on BofA’s assets beyond those assigned to the asset pool. Full-recourse lending is standard practice for the Fed’s normal discount window lending, though some of its Section 13(3) lending has been done on a nonrecourse basis. However, the Fed retained recourse to BofA for the fees and interest on the loan facility (Fed 2009b). The Fed’s credit exposure was limited by its seniority in the overall Asset Guarantee Program (see Figure 1) and loss-sharing arrangement in the loan facility arrangement (insofar as BofA would absorb the first 10% of losses exceeding \$21.1 billion) (Fed 2009a).¹³ The interest on undrawn and

¹¹ Notably, the similar guarantee program for Citi—on which the government modeled the BofA package—specifically excluded derivative assets; see Arnold (2025).
¹² However, because of the 90–10 loss-sharing arrangement, BofA would have to repay 10% of the losses to the Fed, so in that respect, one could say the lending was with partial recourse because of that repayment obligation (GAO 2011). See the Citigroup 2008 annual report, describing this “partial recourse” (the Citi facility featured the same 90–10 loss-sharing policy) (Citi 2009, 45).
¹³ The Fed’s first Section 129 report to Congress on the Asset Guarantee Program also added that the Fed’s “credit exposure would be further limited by Bank of America’s pledge of U.S. Treasury securities or other assets eligible for Reserve Banks to purchase under section 14(b) of the Federal Reserve Act” (Fed 2009a, 5).

drawn amounts also mitigated the Fed's exposure (see Key Design Decision No. 8, Rates and Fees).

Then-General Counsel Alvarez said that even in a scenario where the market value of collateral was deteriorating, with no recourse to BofA (except for interest and fees), the 90–10 loss-sharing agreements were adequate to secure the Fed to its satisfaction given the greater discount to par value that would then be present and the Fed's ability to hold the collateral until it recovered value (Alvarez 2022).

Alvarez said that another measure of security to the Fed's lending was that in the extreme event that it had to seize collateral, the Fed could “perhaps tempter the loss by holding the assets long enough for them to come back in value”; in other words, the Fed could potentially hold the asset pool assets to maturity and receive par value in the unlikely event that it lent and BofA defaulted (Alvarez 2022, 18).

The Fed said that its credit exposure would be “further limited by Bank of America's pledge of U.S. Treasury securities or other assets eligible for Reserve Banks to purchase under section 14(b) of the Federal Reserve Act” (Fed 2009a, 5). Section 14(b) of the FRA authorizes the Fed to deal in “any obligation which is a direct obligation of, or fully guaranteed as to principal and interest by, any agency of the United States” (US Congress 2008a, sec. 14[b]). However, the proposed asset pool ultimately did not include any US Treasury securities and included only a trivial amount of agency residential mortgage-backed securities (see Figure 8 at Appendix) (Glassman et al. 2009). This Section 14(b) reference was absent from the Fed's otherwise similar ring-fencing loan facility balance sheet protection measures for Citigroup (Fed 2008b).

11. Impact on Monetary Policy Transmission: Our research did not uncover any specific impact of the loan facility on monetary policy transmission.

Our research did not uncover any specific impact of the loan facility on monetary policy transmission. As mentioned, the loan was never issued.

12. Other Conditions: While the Fed loan facility did not itself impose conditionality, the broader Asset Guarantee Program imposed conditions on BofA regarding executive compensation and corporate policies.

In connection with the loss-sharing agreements in the Asset Guarantee Program, BofA was required to maintain a foreclosure-mitigation policy in addition to complying with executive compensation and corporate expenditure policies (Fed 2009a). According to the term sheet, “an executive compensation plan, including bonuses, that rewards long-term performance and profitability, with appropriate limitations, must be submitted to, and approved by” the US government (Fed 2009a, 8–9).

Eric Rosengren, then president of the Federal Reserve Bank of Boston, said to Chairman Bernanke that the Fed should consider replacing BofA's management team before adopting the ring-fencing strategy (ultimately, the Fed did not pursue this option) (US House Oversight Committee 2009b).

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Appendix

Figure 7: PIMCO's Breakdown of BofA Proposed Asset Pool, Year-End 2008

Asset Class	Carrying Value
Lending Exposure	
Leveraged Finance	
Transitory ("Hung Exposure") – US	1,587
Leveraged Finance Bridge – US	12
Retained Bond Positions In lieu of Loan	1,454
Relationship – US	327
<i>Subtotal – Leveraged Finance</i>	3,380
Commercial Real Estate	
US	7,363
GPI Portfolio – US	752
<i>Subtotal – Commercial Real Estate</i>	6,115
Total Lending Exposure	11,495

Asset Class	Carrying Value
CDOs	
ABS CDOs (Investment Portfolio)	
U.S. Super Senior – Sub Prime	976
U.S. Super Senior – Non Sub Prime	2,078
Retained Positions – Liquidates CDOs	2,095
<i>Subtotal – ABS CDOs</i>	5,149
ABS CDOs (Trading Portfolio)	
U.S. Super Senior – Unhedged	800
Lone Start Proxy	5,007
<i>Subtotal – ABS CDOs</i>	5,807
Total CDOs	10,956

Asset Class	Carrying Value
Investments	
Investment Portfolio	
Agency RMBS	6
Alt-A RMBS	2,294
CMBS	2,605
Non Resi MBS	661
Non Resi CDO	353
Other	189
Prime RMBS	1,405
Sub-prime RMBS	1,984
Total Investments	9,497

Asset Class	Max Loss Exposure
Derivative Asset and Counterparty Exposure (MTM Receivables)	
Merrill Lynch	
Hedged CDOs (cash and derivative)	3,022
Other Reference Assets	48,633
DPCs – Referenced Assets	17,158
<i>Subtotal – Merrill Lynch</i>	68,813
Bank of America	
U.S. Super Senior – Hedged	1,420
DPCs – Referenced Assets	12,757
<i>Subtotal – Bank of America</i>	14,177
Total Derivate Asset and Counterparty Exposure	82,990

Source: Glassman et al. 2009.

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